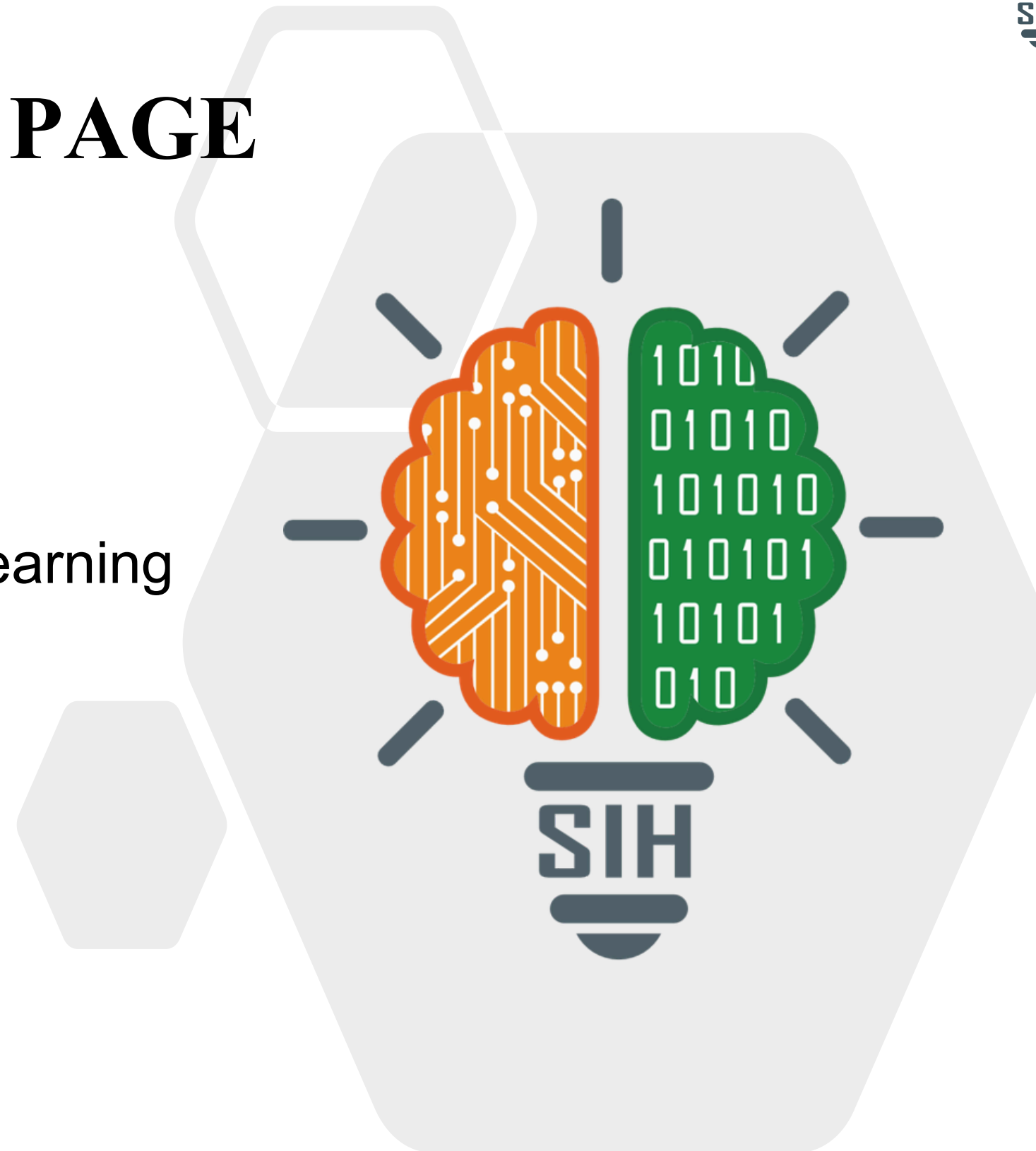


TITLE PAGE

- **Problem Statement ID** – 26205
- **Problem Statement Title-** AI-based learning platform for Neurodivergent learners
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)**

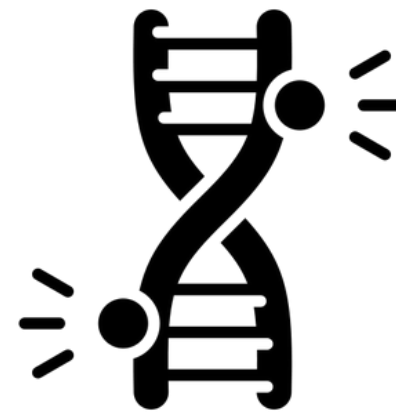


IDEA TITLE

PROBLEM STATEMENT



E-learning Platform based on artificial intelligence created for neurodivergent users (ADHD, Dyslexia, ASD). Adaptation of UI layout, complexity of text, and pace of studying through affective sensing and scaffolding via LLM.



How Does It Solve the Problem?

It gets rid of the inflexible "one-size-fits-all" method that leads to cognitive overload and task abandonment, swaps stress-inducing leaderboards and strict timers for self-paced mastery trees and co-regulation breaks, turns complicated study material into ADHD/dyslexia-friendly visual summaries immediately.

UNIQUENESS



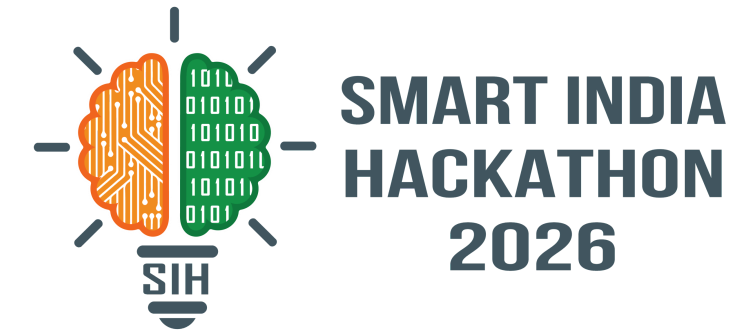
Multimodal Neuro-Adaptation: Combination of affective sensors (gaze tracking, fatigue detection), and text generative AI.



Gamification without Competition: Reward cycles focused on personal development, self-regulation, and maintenance of flow state.

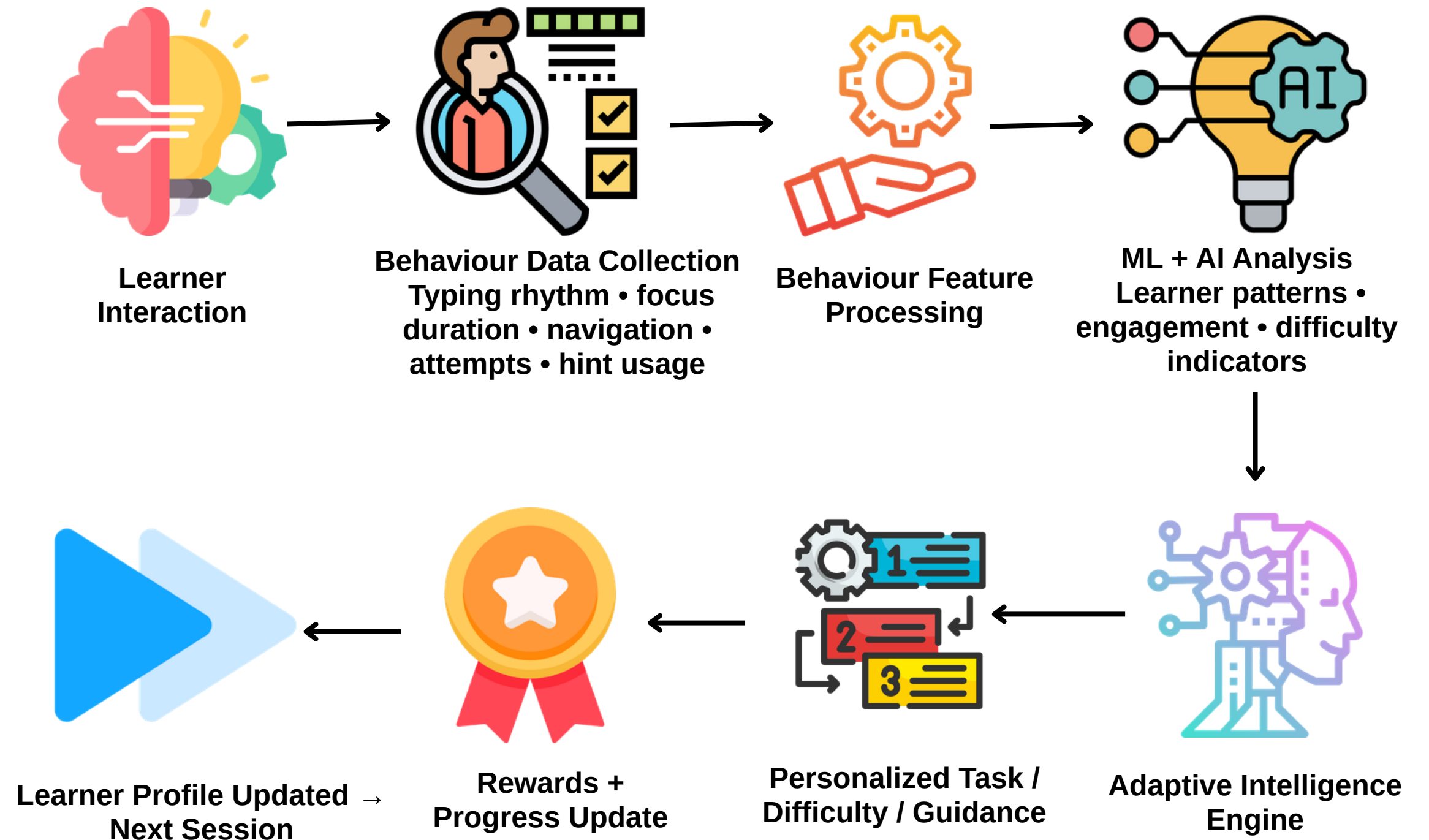
Your Team
Name

TECHNICAL APPROACH

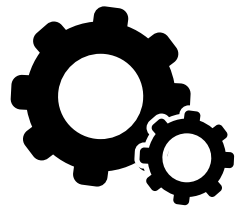


TECH STACK

- **Frontend:** React.js, HTML, CSS, JavaScript
- **Backend:** Node.js + Express.js
- **Database:** MongoDB
- **AI/ML:** Python, FastAPI, Scikit-learn
- **LLM:** Ollama with Llama/Mistral
- **AI Processing:** Behaviour analysis + adaptive decision engine



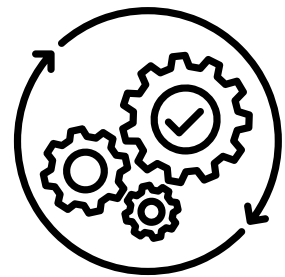
Feasibility Analysis



Feasible through browser-based APIs (TensorFlow.js, Web Speech) and text-simplification pipelines on LLMs.



Economic: Low-cost through client-side processing without the need for expensive cloud servers, allowing for scalable software-as-a-service subscriptions.



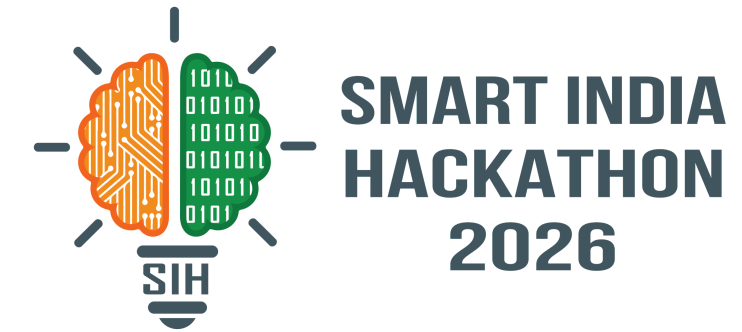
Operational: Integrable to platforms such as Canvas and Moodle using standard LTI.



Market: Market need driven by compliance (Section 508 ADA, UDL) and need for learning tools.

Your Team
Name

IMPACT AND BENEFITS



Impacts



- Personalized learning for every student.
- Increased engagement and motivation.



- Better learning outcomes through adaptive guidance.
- Inclusive education through AI.



- Gamified missions, rewards, badges and streaks encourage consistent participation.
- Breaks complex learning tasks into smaller, manageable missions

Benefits

LEARNERS:

- Real-time guidance
- Personalized support
- Adaptive learning experience
- Improved engagement

Educators / Institutions:

- Progress insights
- Behavior insights
- Better understanding of learner needs

EdTech Platforms:

- Integration of the adaptive AI engine through API.
- Ability to provide personalized learning at scale.
- Free, premium and institution plans support different adoption levels.

1. Deckker & Sumanasekara, Systematic Review on AI in Special Education: Enhancing Learning for Neurodiverse Students, 2025
2. Sheejamol P. T.; Anu Mary Chacko; S. D. Madhu Kumar, Beyond the One-Size-Fits-All: A Systematic Review of Personalized and Gamified e-Learning for Neurodivergent Learners, 2025
3. Enhancing Adaptive Personalized Learning Interfaces with Generative AI for Individuals with ADHD, Proceedings of the 16th International Conference of HCI Design & Research (ACM), 2025
4. Halkiopoulous & Gkintoni, A Systematic Review of Personalized and Gamified e-Learning for Neurodivergent Learners: Cognitive and Engagement Perspectives, 2024–2025
5. Challenging Cognitive Load Theory: The Role of Educational Neuroscience and Artificial Intelligence in Redefining Learning Efficacy, 2024
6. Gamifying Learning with AI: A Pathway to 21st-Century Skills, Early Child Development and Care journal, 2024

